

Christopher P. Domingos
Site Vice President

DTE Electric Company
6400 N. Dixie Highway, Newport, MI 48166
Tel: 734.586.5025 Fax: 734.586.5295
Email: christopher.domingos@dteenergy.com



August 14, 2025
NRC-25-0037

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2025-001-00

Pursuant to 10 Code of Federal Regulations (CFR) 50.73(a)(2)(i)(B) and 50.73(a)(2)(v)(D), DTE Electric Company (DTE) is submitting LER No. 2025-001-00, "Div. 1 Transformer 64 Load Tap Changer Failure Potentially Prohibited by Technical Specifications and Potential Loss of Safety Function".

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

A handwritten signature in blue ink, appearing to read "Christopher P. Domingos", with a long horizontal flourish extending to the right.

Christopher P. Domingos
Site Vice President

Enclosure: Licensee Event Report No. 2025-001-00, "Div. 1 Transformer 64 Load Tap Changer Failure Potentially Prohibited by Technical Specifications and Potential Loss of Safety Function"

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

**Enclosure to
NRC-25-0037**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2025-001-00
“Div. 1 Transformer 64 Load Tap Changer Failure Potentially Prohibited by Technical
Specifications and Potential Loss of Safety Function”**



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk aid: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Fermi 2

2. Docket Number

05000341

3. Page

1 OF 3

4. Title

Division I Transformer 64 Load Tap Changer Failure Potentially Prohibited by Technical Specifications and Potential Loss of Safety Function

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
06	17	2025	2025	- 001 -	00	08	13	2025	N/A	☐ 052	00

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ Other (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Eric Frank – Manager, Nuclear Licensing

Phone Number (Include Area Code)

734-586-4772

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
X	EB	TTC	G080	Y					

14. Supplemental Report Expected

☐ No ☒ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year
10	03	2025

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 6/17/25 around 1755 Eastern Prevailing Time (EPT), Station Service Transformer (SST) 64 low voltage alarm was received on Division I Bus after starting Circulating Water Pump #4. SST 64 Load Tap Changer (LTC) was found to be non-functional requiring manual operation of the LTC to be initiated. Subsequent diagnostics revealed a failed time-delay closing relay in the LTC circuitry, which was replaced on 6/19/25 at 1932 EPT. After reviewing historical data, the LTC was last found to be functioning on 2/7/2024. The non-functional LTC could potentially result in lower than minimum required voltages during a postulated loss of coolant accident coincident with degraded voltage conditions.

This Licensee Event Report (LER) is being made under the corresponding requirement in Title 10 of the Code of Federal Regulations (CFR) 50.73(a)(2)(i)(B), due to a condition that is potentially prohibited by Technical Specifications. This LER is also being made pursuant to 10 CFR 50.73(a)(2)(v)(D) as a potential condition that could have prevented the fulfillment of the safety function needed to mitigate the consequences of an accident. Further investigation is being conducted to determine the extent of the condition, and a supplement is planned. The event had no significant effect on public health and safety.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	05000-341	2025	- 001	- 00

NARRATIVE**INITIAL PLANT CONDITIONS**

MODE – 1

Reactor Power – 100

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On 6/17/25 around 1755 Eastern Prevailing Time (EPT), Station Service Transformer (SST) [XFMR] 64, Maximo Location Number R1200S002, low voltage alarm was received on Division I Bus [EB] [BU] after starting Circulating Water Pump #4 [KA][P]. SST 64, Load Tap Changer (LTC) [TTC] (General Electric, Model LR-65) was found to be non-functional requiring manual operation of the LTC to be initiated. The LTC was manually adjusted, which cleared the alarm and raised the bus voltage to 118 V. The LTC was then returned to automatic mode. SST 64 voltage was recorded hourly to ensure voltage was maintained within the operability band (122-to-124-meter volts) and adjusted as necessary to maintain the band. The transformer was confirmed to be operable after manual voltage control was maintained within the required band. Subsequent diagnostics revealed a failed time-delay closing relay [TDC] in the LTC circuitry, which was replaced and tested on 6/19/25 at 1932 EPT. After reviewing historical data, the LTC was last found to be working on 2/7/2024.

The primary function of SST 64 is to provide a means to step down voltage (13.8 kV) from Bus 11 to 4160V for distribution to Division 1 Engineered Safety Feature (ESF) (class 1E power requirements) and Balance of Plant loads. Operability is based on the ability to supply power from the 120 kV switchyard, through SST 64, to 4160V buses 64B and 64C. This transformer is also sized to accommodate the simultaneous starting of all ESF loads on receipt of an accident signal, without the need for load sequencing. The LTC modulates the voltage on the transformer to keep a nominal 121.8 meter-Volts to ensure Division 1 offsite source has adequate voltage to support the startup and subsequent functionality of the Fermi 2, Division 1 ESF loads without tripping during a Loss of Offsite Coolant Accident (LOCA) transient.

While SST 64, LTC control switch was set to automatic, would not automatically tap to control voltage. While the LTC is in manual mode, procedure states that if the voltage range of 122-to-124-meter volts is not met, then Buses 64B and 64C are considered INOPERABLE. The basis of this requirement is the offsite circuit cannot be lost (i.e. by actuation of the degraded voltage relaying) during a design basis accident and must be capable of starting all required ESF loads on the receipt of an accident signal, so this condition impacts the ability to meet the requirements of Technical Specification (TS) 3.8.1.

TS 3.8.1 “AC Sources – Operating” requires (During MODES 1,2 and 3) two qualified circuits between the offsite transmission network and the on-site Class 1E AC Electrical Power Distribution System and two emergency diesel generators (EDGs) per division. When one offsite circuit is inoperable, then CONDITION D is entered to restore the offsite circuit to OPERABLE status within 72 hours. If the REQUIRED ACTION and COMPLETION TIME is not met, MODE 3 needs to be entered within 12 hours. Further investigation is being done to determine when the voltage level was outside of the voltage band exceeding the completion time of 72 hours along with determining whether the condition was a potential loss of a safety function. The results from the investigation will be provided in a supplemental report.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

Further investigation is being conducted for the safety consequences and implications and will be supplemented.

**LICENSEE EVENT REPORT (LER)
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There were no safety consequences or radiological releases associated with this event. At no time during this event was there potential for endangering the public health and safety.

CAUSE OF THE EVENT

The LTC failed due to a failed time-delay relay within the LTC controls.

CORRECTIVE ACTIONS

Diagnosis identified a failed Time Delay Relay. LTC was placed back in AUTO mode following replacement of the relay. Actions to address procedural and human performance gaps will be addressed in the Corrective Action Program.

PREVIOUS OCCURRENCES

An examination of Condition Reports from the past three years yielded the following information:

CR-2024-39536 - On 5/13/24, SST 64 LTC did not maintain voltage automatically. Operators manually increased the voltage from approximately 119V to 121V. Although the transformer continued to supply proper voltage and frequency to essential safety buses, a work order was requested to investigate and repair the issue. However, this work was never performed until the alarm in June 2025.